



Elettra Sincrotrone Trieste

MXCuBE site report

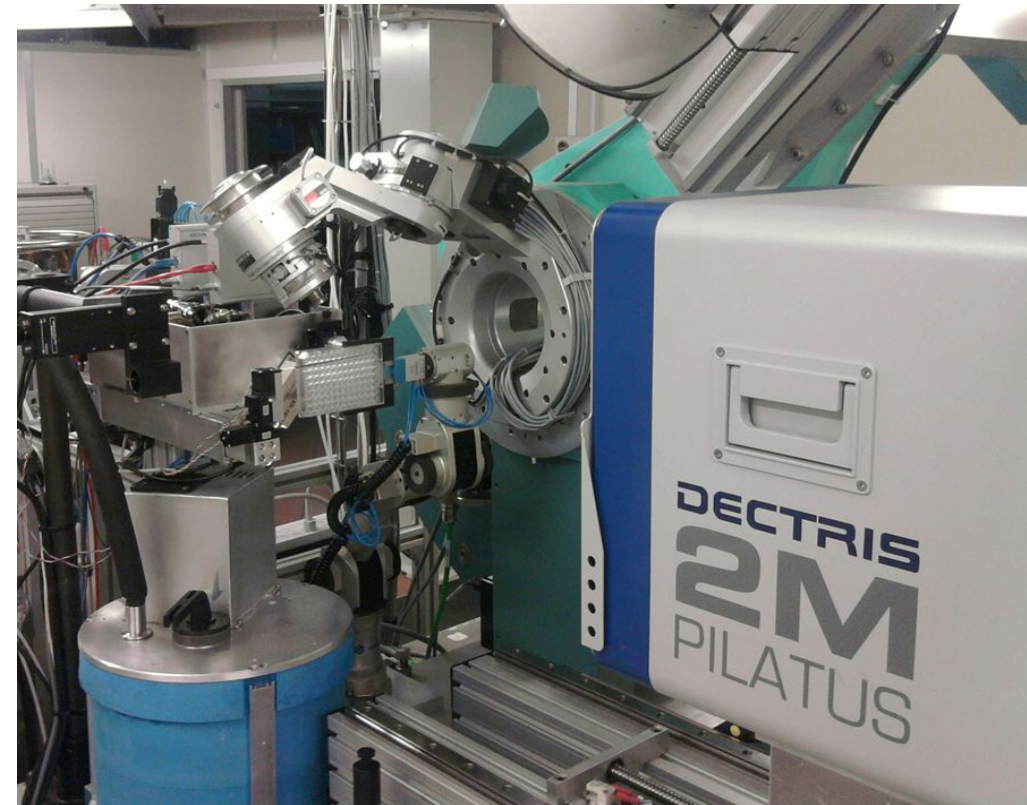
Elettra Sincrotrone Trieste

Brief beamlines summary (1)

XRD1 (X-ray diffraction):

General purpose XRD - multiple sample supports accepted

- Huber Kappa Goniometer
- Pilatus 2M
- Arinax BZoom

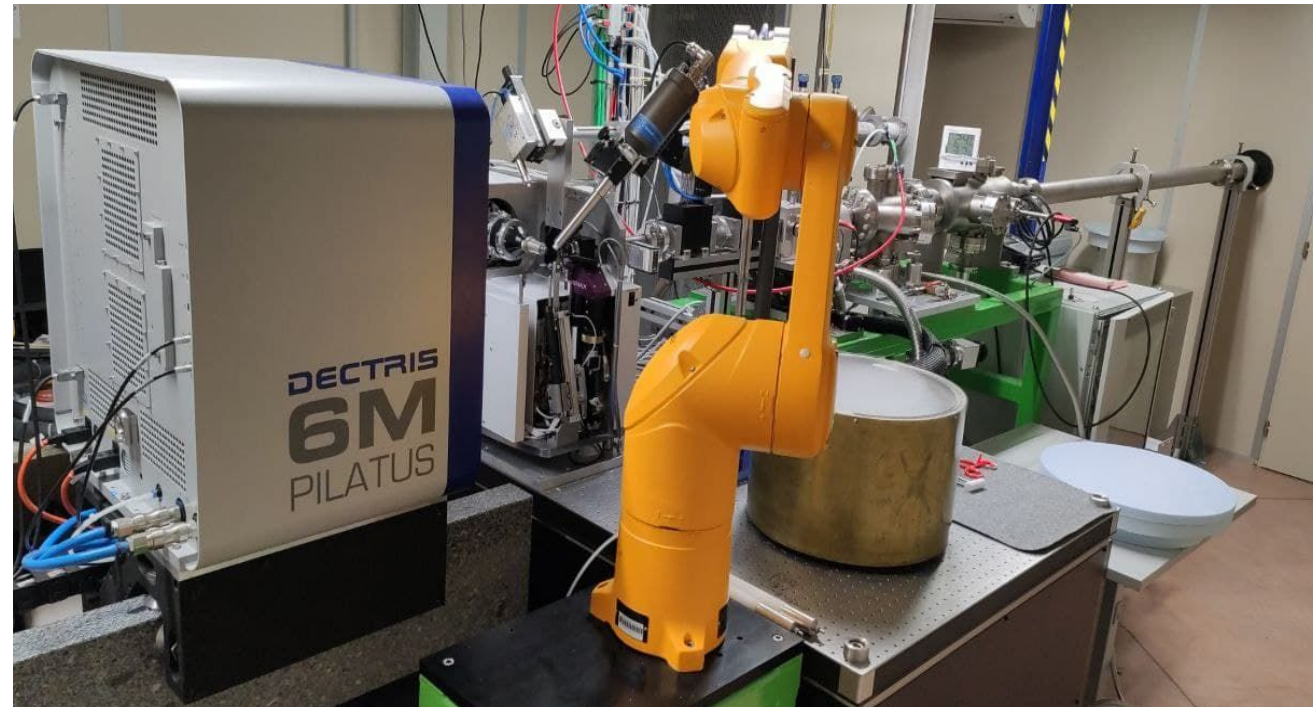


Brief beamlines summary (2)

XRD2 (X-ray diffraction 2):

Focussed on MX - SPINE standard samples only

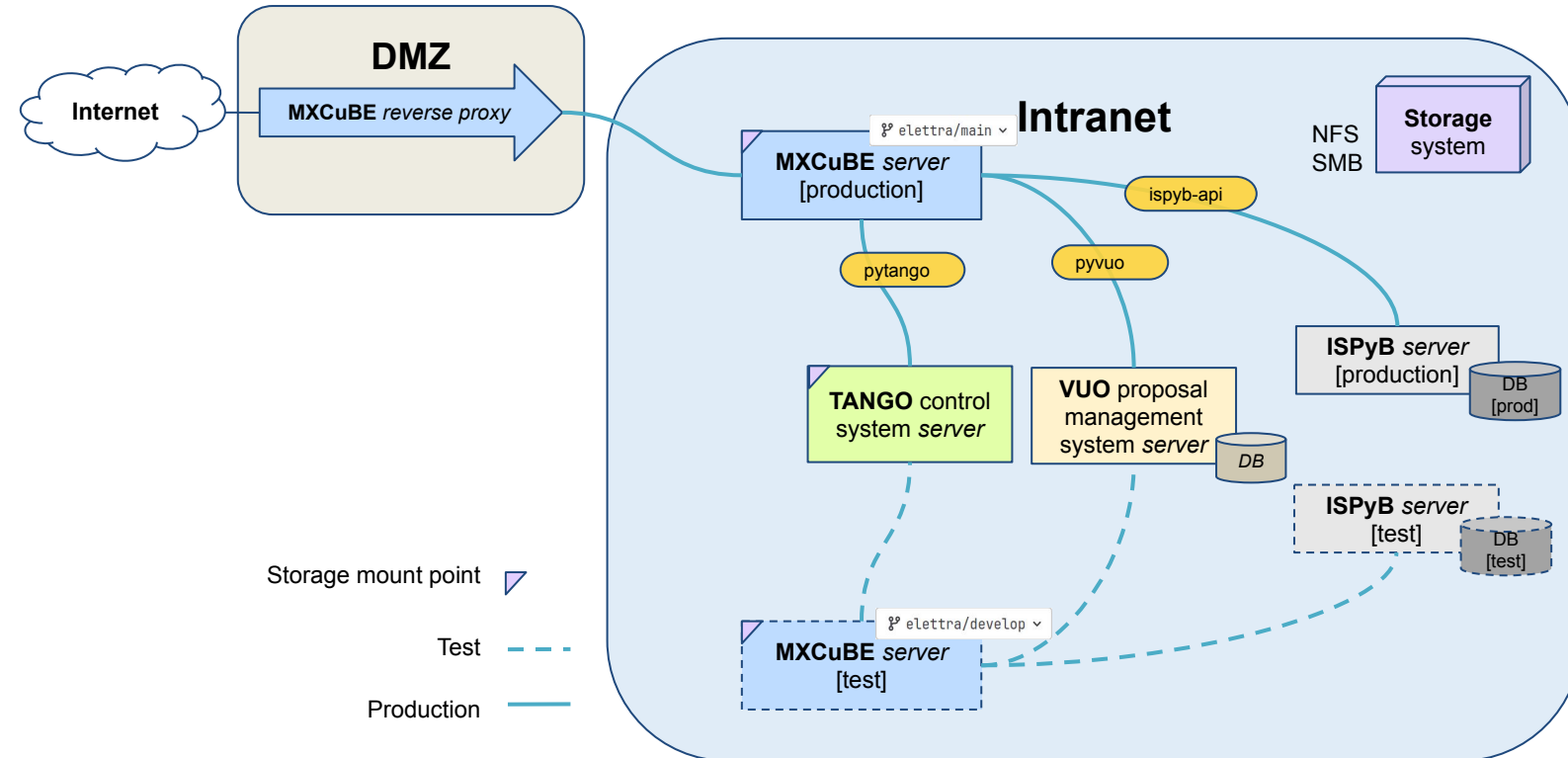
- Arinax MD2s
- Pilatus 6M
- Arinax BZoom
- Custom sample changer
(Staubli TX60L on internally
developed 12 Unipucks dewar
and in-house developed
Val3 code)



MXCuBE status (1)

XRD1

- Deployment at glance:
 - Rocky Linux 9 (Red Hat):
 - ~~Python 3.9~~ → Python 3.10
 - Virtual environment
 - ~~pip~~ → poetry 2.0.1
 - MXCuBE Web version 4
- Open to the internet



MXCuBE status (2)

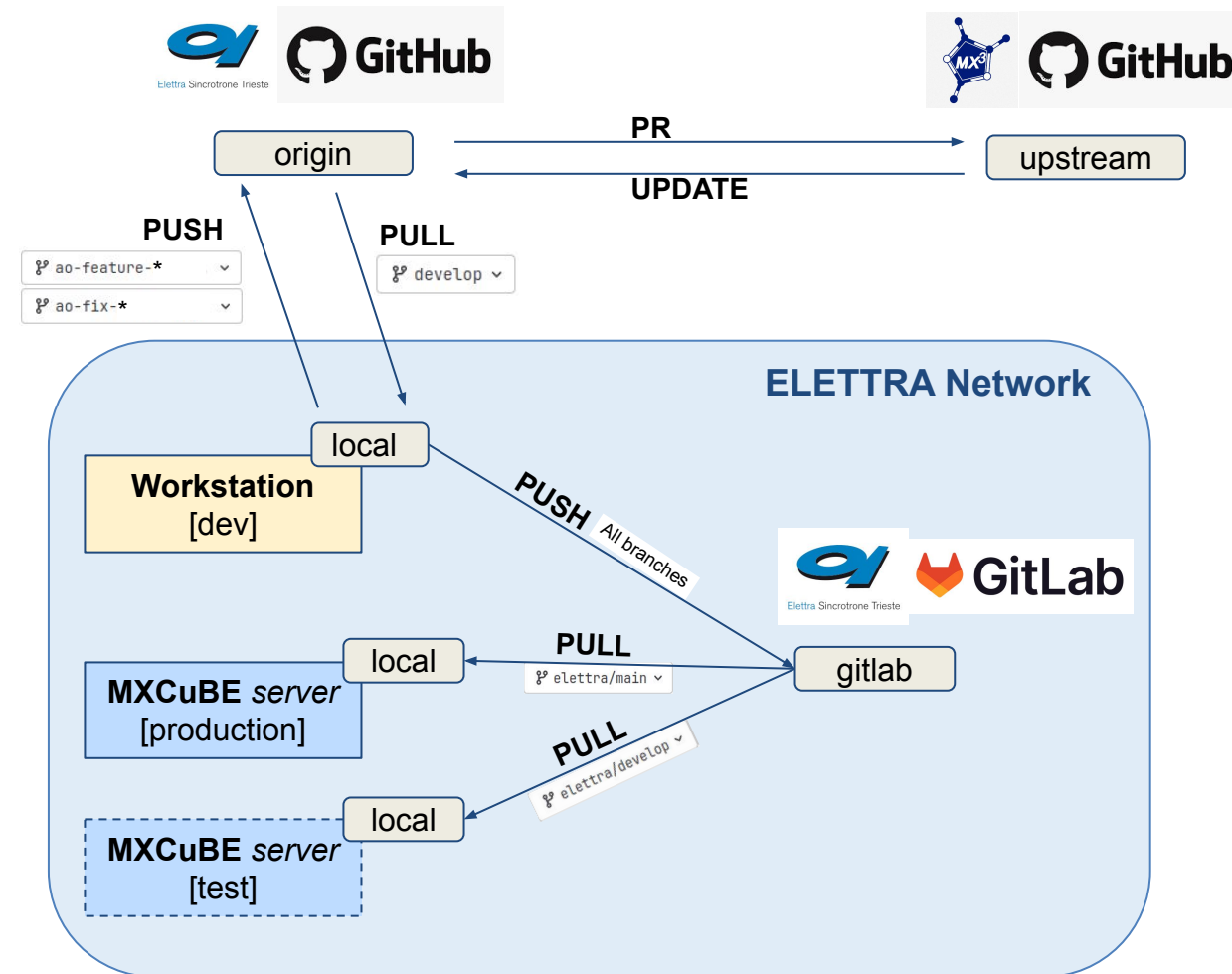
XRD2

- Deployment at glance:
 - Ubuntu 18.04.4 LTS
 - Python 2.7
 - pip
 - MXCuBE Web version 3
- New features developed for the MXCuBE web version 3
- Porting from MXCuBE 3 to MXCuBE 4 is still on going

Software Lifecycle (1)

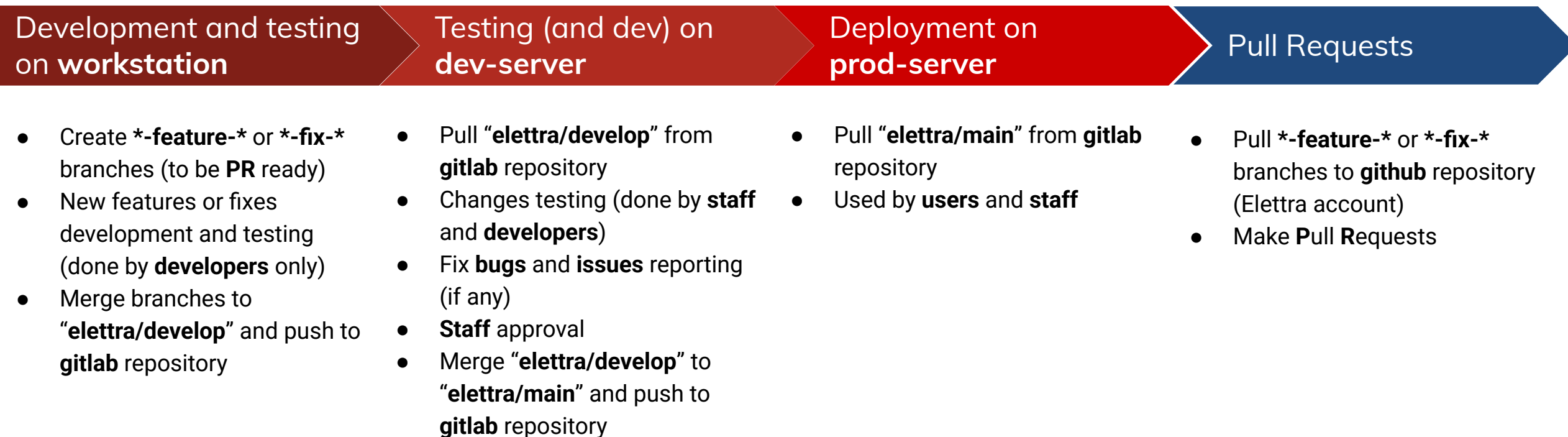
Versioning:

- GitHub Elettra public account
 - <https://github.com/ELETTA-Sincrotrone-Trieste/mxcubecore>
 - <https://github.com/ELETTA-Sincrotrone-Trieste/mxcubeweb>
 - Self-hosted Gitlab
 - Branching as suggested in the MXCuBE contribution guidelines and two more branches “elettra/develop” and “elettra/main”



Software Lifecycle (2)

Dev, Test and Deploy:



Pull Requests (1)

Branched from a **develop** commit of the last year

Mxcubecore branches

```
ao-feature-add_detector_modes_to_acq_params
ao-feature-add_elettra_hwo
ao-feature-add_space_groups_to_acq_params
ao-feature-custom_centring_name
ao-feature-diffr_is_moving
ao-feature-hwo_allow_setter_decorator
ao-feature-hwo_logging_decorator
ao-feature-lims_model
ao-feature-opt_phase_ctrl
ao-fix-action_abort
ao-fix-common_model
ao-fix-crash_manual_samp_after_clear
ao-fix-make_olog_consistent_with_ui
ao-fix-poller_old_value_reset
ao-fix-stop_stream_err_handling
ao-fix-tango_poll_restart_after_err
ao-fix-tango_poller_exceptions_handling
ao-fix-user_lvl_log
```

Mxcubeweb branches

```
ao-feature-alert_if_no_sessions
ao-feature-avoid_negative_clicks
ao-feature-avoid_user_with_no_sessions
ao-feature-change_default_centring_method
ao-feature-clean_user_db_on_startup
ao-feature-conf_mxcube_pid
ao-feature-custom_centring_name
ao-feature-disable_add_tasks_if_no_samp
ao-feature-disable_auto_centring_option_dynamically
ao-feature-disable_automount_option_dynamically
ao-feature-disable_get_samp_frm_samp_chng
ao-feature-filtr_proposals_with_no_sessions
ao-feature-hide_space_group_dynamic
ao-feature-invert_sw_bg_color
ao-feature-keep_video_size_after_refresh
ao-feature-local_host_by_domain
ao-feature-make_detector_modes_conf_from_srv
ao-feature-make_tasks_menu_dynamic
ao-feature-manual_samp_ui_config
ao-feature-opt_phase_ctrl
ao-feature-opt_samp_chng
```

```
ao-feature-optional_uiprops
ao-feature-synch_manual_samp_with_lims
ao-feature-ui_custom_motors
ao-feature-xrf_spectrum_opt
ao-fix-apply_mesh_parameters
ao-fix-beamline_action_abort
ao-fix-check_session_existence
ao-fix-compare_osc_in_data_collection
ao-fix-configmodel_stream_dflt
ao-fix-dc_duplication_on_sample_mounting
ao-fix-dc_subdir_validation
ao-fix-log_frontend_to_backend
ao-fix-mov_state_aperture
ao-fix-resize_video_stream
ao-fix-samp_chng_offline
ao-fix-sample_cntrls_centered
ao-fix-sequential_synch_with_lims
ao-fix-task_start_osc_compare
ao-fix-typo_sampleview
ao-fix-unmount_manual_samp
ao-fix-video_stream_chng_dim
```

Pull Requests (2)

Branched from a **develop** commit of the last year

Mxcubecore branches

```

ao-feature-add_detector_modes_to_acq_params
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ao-feature-opt_phase_ctrl
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ao-fix-crash_manual_samp_after_clear
ao-fix-make_vlog_consistent_with_ui PR
ao-fix-poller_old_value_reset
ao-fix-stop_stream_err_handling
ao-fix-tango_poll_restart_after_err
ao-fix-tango_poller_exceptions_handling PR
ao-fix-user_lvl_log

```

Mxcubeweb branches

```

ao-feature-alert_if_no_sessions
ao-feature-avoid_negative_clicks PR
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ao-feature-change_default_centring_method
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ao-feature-keep_video_size_after_refresh
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ao-feature-opt_phase_ctrl
ao-feature-opt_samp_chng

```

```

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ao-fix-sample_cntrls_centered
ao-fix-sequential_synch_with_lims
ao-fix-task_start_osc_compare PR
ao-fix-typo_sampleview
ao-fix-unmount_manual_samp
ao-fix-video_stream_chng_dim

```

Pull Requests (3)

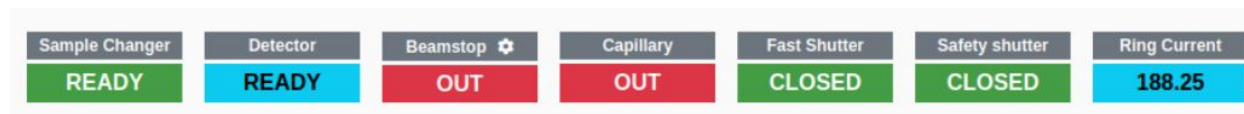
Some branches

- **mxcube**core:

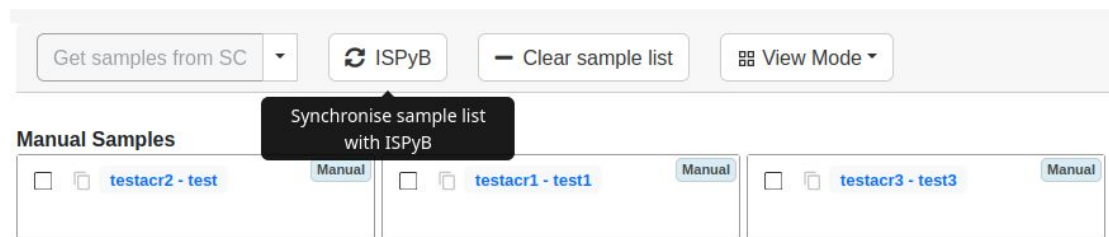
- ao-fix-tango_poller_exceptions_handling ⇒ [PR #1218](#)
- ao-feature-hwo_logging_decorator ⇒ [PR #1223](#)

- **mxcube**web:

- ao-feature-avoid_negative_clicks ⇒ [PR #1649](#)
- ao-feature-invert_sw_bg_color ⇒ [PR #1645](#)



- ao-feature-synch_manual_samp_with_lims ⇒ PR not yet



```
from mxcube.core.HardwareObjects.mockup.ActuatorMockup import ActuatorMockup
from mxcube import hwo_header_log

...

class MotorMockup(ActuatorMockup, AbstractMotor):

    ...

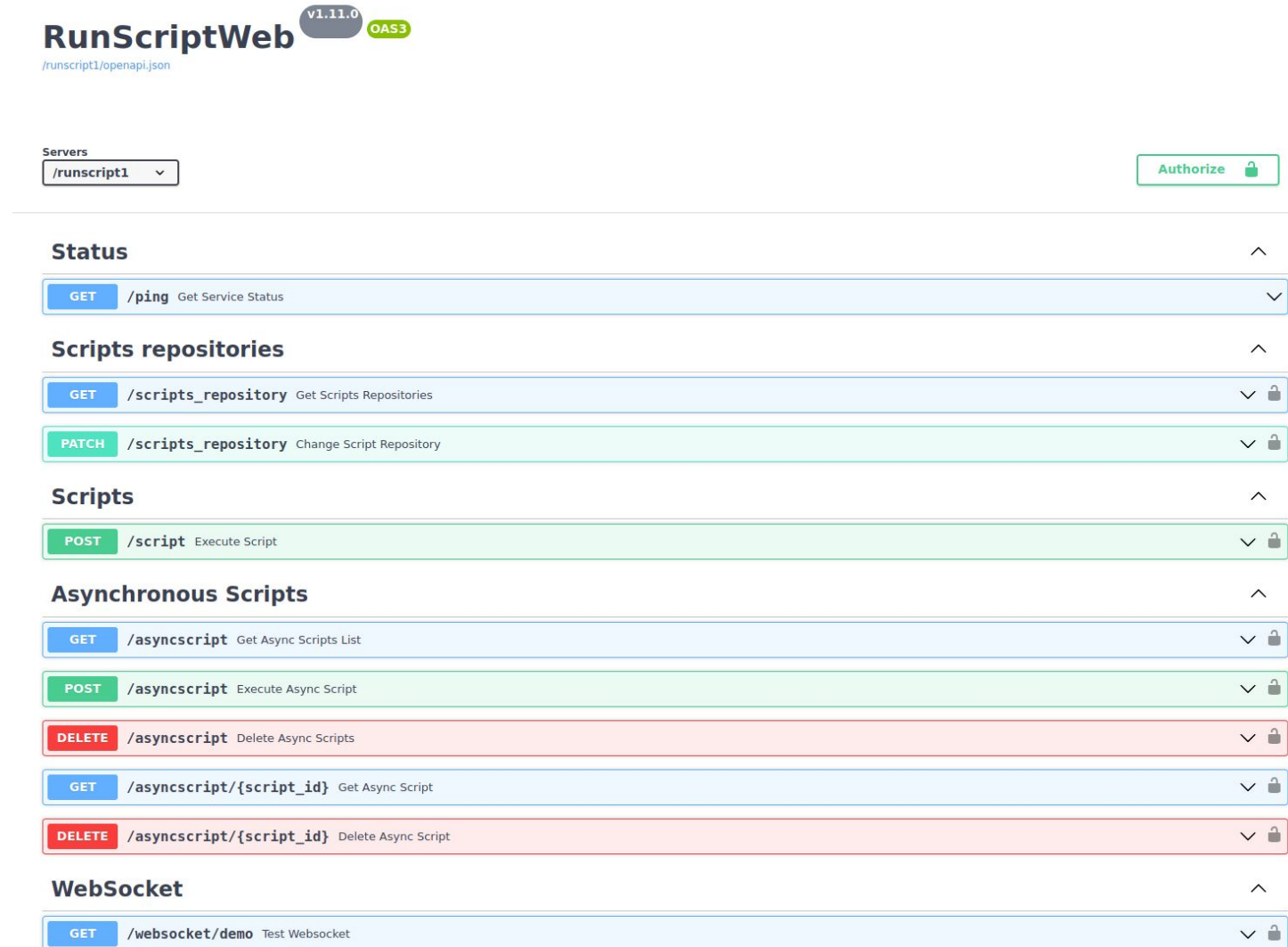
    @hwo_header_log
    def _move(self, value):
        """Simulated motor movement.
        Args:
            value (float): target position
        """
        ...
```

2025-05-16 18:54:29,881 |HWR.MotorMockup|DEBUG | (kappa) In _move(11.1)

RunScriptWeb (1)

The service

- Executes scripts on remote machine
- Accessible via a REST API
- Python and based on ⚡ FastAPI
- Developed at ELETTRA (replace Tango's role in the storage system) simple, easy to integrate with VUO
- Supports two levels of authorization
- Script execution restricted to a specified Git repository
- Synchronous and asynchronous scripts execution
- Websocket
- RSW python client



The screenshot shows the OpenAPI documentation for RunScriptWeb v1.11.0. The interface includes a 'Servers' dropdown set to '/runscript1' and an 'Authorize' button. The API endpoints are organized into sections: Status, Scripts repositories, Scripts, Asynchronous Scripts, and WebSocket. Each endpoint is listed with its HTTP method, path, and description.

Method	Path	Description
GET	/ping	Get Service Status
GET	/scripts_repository	Get Scripts Repositories
PATCH	/scripts_repository	Change Script Repository
POST	/script	Execute Script
GET	/asyncscript	Get Async Scripts List
POST	/asyncscript	Execute Async Script
DELETE	/asyncscript	Delete Async Scripts
GET	/asyncscript/{script_id}	Get Async Script
DELETE	/asyncscript/{script_id}	Delete Async Script
GET	/websocket/demo	Test Websocket

RunScriptWeb (2)

Process pipeline:

- 2 HPC nodes:
 - CPU: 2 x Intel Xeon Gold 6248R@3.00GHz
 - RAM: 768 GB
 - NET: 25 Gb/s Ethernet
 - OS: Rocky Linux 9
- RSW is called from MXCuBE
- Autoproc, fast_dp, adxv, etc

```
curl -X 'POST' \
  'https://xrd2-process.elettra.eu/runscript1/asynscript?use_bash=false&remove_on_exit=false&use_tty=false&uid=0&gid=0&legacy=false&scripts_repository=rsw-xrd2-op_scripts' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer d0nYPTZqQ1Q7U2ihRsEK' \
  -H 'Content-Type: application/json' \
  -d '{
    "command": "do_sleep_op.sh",
    "id": "test_id"
  }'
```

```
curl -X 'GET' \
  'https://xrd2-process.elettra.eu/runscript1/asynscript/test_id?legacy=false' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer d0nYPTZqQ1Q7U2ihRsEK'
```

```
{
  "command": "/scripts/rsw-xrd2-op_scripts/do_sleep_op.sh",
  "id": "test_id",
  "pid": 569397,
  "status": "EXITED",
  "ret_code": 0,
  "stdout": "-- buonanotte -- originale\nSCRATCHDIR=/net/xrd2-ppu/dataInt -\ne\n",
  "stderr": "",
  "creation_date": "2025-05-18T15:47:01.725628",
  "start_date": "2025-05-18T15:47:01.120000",
  "end_date": "2025-05-18T15:47:10.742995",
  "uid": 0,
  "gid": 0
}
```

op_scripts

master op_scripts / +

History Find file Edit Code

DIMPLE-->DUMBS pipelines must accept xds4elettra input
Your Name authored 1 hour ago cf5b1136

Name	Last commit	Last update
xds4ilo	added xds only pipeline mostly 4 ILO p...	1 year ago
README.md	fixes for RLP attachments	3 years ago
do_adxv.sh	Massive update for _wdg_ stuff + BEST ...	1 month ago
do_adxv_op.sh	Dimple attachments fixes	2 years ago
do_autoproc.sh	Made aP aware of XDS_ASCII.HKL posit...	4 weeks ago
do_autoprocpar.sh	Made aP aware of XDS_ASCII.HKL posit...	4 weeks ago
do_autosoLop.sh	Don't trust ISPyB number of frames/DC...	1 month ago
do_buster_op.sh	Don't trust ISPyB number of frames/DC...	1 month ago
do_catxml.sh	Massive update for _wdg_ stuff + BEST ...	1 month ago
do_dialsrip_op.sh	Made aP aware of XDS_ASCII.HKL posit...	4 weeks ago
do_dimple_op.sh	DIMPLE-->DUMBS pipelines must acce...	1 hour ago
do_dozor.sh	Massive update for _wdg_ stuff + BEST ...	1 month ago

```
{
  "command": "/scripts/rsw-xrd2-op_scripts/do_sleep_op.sh",
  "id": "test_id",
  "pid": 569397,
  "status": "RUNNING",
  "ret_code": null,
  "stdout": null,
  "stderr": null,
  "creation_date": "2025-05-18T15:47:01.725628",
  "start_date": "2025-05-18T15:47:01.120000",
  "end_date": null,
  "uid": 0,
  "gid": 0
}
```

Plans for the next six months

- Update MXCuBE with the latest version from the upstream repository
- Go on with the Pull requests
- Complete porting from MXCuBE v3 to v4 for **XRD2**
- **Dark Period**

